# Chapter 18 Hunter 16 Interrupts

This chapter briefly describes the **Hunter 16** interrupts.

The BIOS interrupts are described in detail in Chapter 19 to Chapter 31. The hardware interrupts are described in section 32.3.

The MS-DOS interrupts are described in the *MS-DOS Programmer's Manual*. Examples illustrating how to use the BIOS interrupts are given later, in the chapters on the individual subsystems.

## 18.1 The interrupts

The 8088 microprocessor supports 256 different interrupts. There three types:

- **BIOS interrupts**, instructions executed by programs, as an economical way of calling routines in the BIOS (such as the video display or communications services).
- **Hardware interrupts**, issued by a device manipulating control lines, as a way of requesting service from the BIOS (such as responding to a timer tick or key depression).
- **Microprocessor interrupts**, issued by the microprocessor itself under certain circumstances (such as an attempt to divide by 0 or arithmetic overflow)

All interrupts are vectored from low RAM, to allow an application to service the interrupt with its own routine. The routine can be used to change the normal response to an interrupt, or to provide a response where one is not normally made (as indicated by IRET in the following table). The vectors are restored to their defaults by a cold start (power up in non-resume mode, or after emergency breakout or loss of all power).

The main vector addresses and default contents on **Hunter 16** are summarised in the following table. To change a vector, you can write direct to memory or use the DOS 'set interrupt vector function (see the *MS-DOS Programmer's Manual*).

**WARNING:** You must not change the vector for INT 2 (which services NMIs), as NMIs are vital to operation of **Hunter 16**. If you change this vector, **Hunter 16** will stop operating.

| Address | INT | Description | Default Vector |
|---|---|---|---|
| 00..03H | 00H | Divide by 0 | IRET |
| 04..07H | 01H | Single step | IRET |
| 08..0BH | 02H | NMI | NMI routine. Do not alter! |
| 0C..0FH | 03H | Breakpoint | IRET |
| 10..13H | 04H | Arithmetic overflow | IRET |
| 14..17H | 05H | Print screen | BIOS print screen |
| 18..1BH | 06H | RESERVED | IRET |
| 1C..1FH | 07H | FPO instruction | IRET |
| 20..23H | 08H | H/W. Time of day | BIOS timer tick routine |
| 24..27H | 09H | H/W. Keyboard | Key store routine |
| 28..2BH | 0AH | RESERVED | IRET |
| 2C..2FH | 0BH | H/W. COM2 port | IRET |
| 30..33H | 0CH | H/W. COM1 port | IRET |
| 34..37H | 0DH | H/W. Fixed disk | IRET |
| 38..3BH | 0EH | H/W. Diskette | Internal UARTs |
| 3C..3FH | 0FH | H/W. Printer | IRET |
| 40..43H | 10H | Screen control | BIOS video routine |
| 44..47H | 11H | Equipment check | BIOS equipment check routine |
| 48..4BH | 12H | System memory | BIOS memory size routine |
| 4C..4FH | 13H | Disk control | BIOS disk control routine |

| Address | INT | Description | Default Vector |
|---|---|---|---|
| 50..53H | 14H | Communications | BIOS comms routine |
| 54..57H | 15H | Cassette control | BIOS extension routine |
| 58..5BH | 16H | Keyboard control | BIOS keyboard control routine |
| 5C..5FH | 17H | Printer | BIOS printer routine |
| 60..63H | 18H | Not implemented | IRET |
| 64..67H | 19H | Bootstrap | BIOS disk bootstrap |
| 68..6BH | 1AH | Time of day | BIOS clock/RTC routine |
| 6C..6FH | 1BH | [CTRL][Break] | IRET |
| 70..73H | 1CH | Timer 'tick' | IRET |
| 74..77H | 1DH | Video parameters | BIOS video parameters |
| 78..7BH | 1EH | Diskette parameters | BIOS diskette parameters |
| 7C..7FH | 1FH | Graphics characters | 0:0000 |
| 80..9CH | 20H ..27H | MS-DOS services | See the *MS-DOS Programmer's Manual* |
| 19C..19FH | 67H | Expanded memory | BIOS EMS memory |
| 1C0..1C3H | 70H | Comms timeout | IRET |
| 1C4..1C7H | 71H | Barcode input | Barcode device driver (option) |
| 1C8..1CBH | 72H | Power up/down/reset | Disabled |

**NOTE:** To use the INT 72H vector, you must also use INT 15H function 87H to link to your routine.

## 18.2 BIOS interrupts

BIOS interrupts provide convenient access to low-level facilities. The following table summarises the BIOS interrupts provided on **Hunter 16**. These are described more fully in the remainder of this chapter. (Example routines illustrating their use are also provided in the descriptions of the individual subsystems, later in this guide.)

The Hunter 16 BIOS interrupts are entirely compatible with the IBM PC-XT BIOS, except where the following symbols appear:

- **+** means this interrupt provides facilities in addition to those of the equivalent PC BIOS interrupt. Using these will make better use of **Hunter 16's** potential, but may reduce portability of your programs.
- **-** means this interrupt does not provide all the facilities of the equivalent PC BIOS interrupt. If you wish to use this interrupt, take careful note of the description of the differences.

If an interrupt has neither of these symbols against it in the following table, you may safely assume that it behaves as in the IBM BIOS and need not read the description included here.

**NOTE:** Many of the extra BIOS interrupts are used to change **Hunter 16's** behaviour, by altering default settings such as the battery type or keyboard layout. These changes are in general preserved until the next cold start. A cold start is produced by a power-up in non-resume mode, or after emergency breakout or loss of all power.

| INT | Standard Function | Differences |
|---|---|---|
| 05H | Print (virtual) screen | + |
| 10H | Screen control | + - |
| 11H | Determine system equipment | |
| 12H | Determine system memory | |
| 13H | Disk control | - |
| 14H | Serial port control | + - |
| 15H | BIOS extension function | + |
| 16H | Keyboard control | + |
| 17H | Printer control | |
| 18H | Activate ROM BASIC/end program | Not implemented |
| 19H | Boot DOS | |
| 1AH | Timer/clock control | + |
| 67H | Expanded (LIM) memory control | + |
| 71H | Barcode input | + |